

Laboratory Exercise :

B. Problem Statements :

1. company wants to develop a simple calculator program that performs division of two numbers.

- If the user enters 0 as denominator, it should throw an ArithmeticException and display a proper error message.
 - The program should also handle any invalid input entered by the user (e.g., entering a string instead of a number) using InputMismatchException.
 - Regardless of an exception, the program should display a message that the calculation attempt has finished.
- CODE (Student.java):

CODE

```
import java.util.InputMismatchException;
import java.util.Scanner;

public class SimpleCalculator {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        try {
            System.out.print("Enter numerator: ");
            int numerator = scanner.nextInt();
            System.out.print("Enter denominator: ");
            int denominator = scanner.nextInt();

            int result = numerator / denominator;
            System.out.println("Result: " + result);
        } catch (ArithmeticException e) {
            System.out.println("Error: Division by zero is not allowed!");
        } catch (InputMismatchException e) {
            System.out.println("Error: Please enter valid integer numbers!");
        } finally {
            System.out.println("Calculation attempt finished.");
            scanner.close();
        }
    }
}
```

```
}
```

OUTPUT :

```
PS C:\Users\Student\Aryan_1026\Exp 7> javac .\SimpleCalculator.java
PS C:\Users\Student\Aryan_1026\Exp 7> java SimpleCalculator
Enter numerator: 10
Enter denominator: 0
Error: Division by zero is not allowed!
Calculation attempt finished.
PS C:\Users\Student\Aryan_1026\Exp 7>
```

2. Write a Java Program to calculate the Result. Result should consist of name, seatno, date, centre number and marks of sem-2 examination. Create a user defined exception class MarksOutOfBoundsException, If Entered marks of any subject is greater than 100 or less than 0, and then program should create a user defined Exception of type MarksOutOfBoundsException and must have a provision to handle it.

CODE :

```
import java.util.Scanner;

// Custom exception class
class MarksOutOfBoundsException extends Exception {
    public MarksOutOfBoundsException(String message) {
        super(message);
    }
}

public class ResultCalculation {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        try {
            System.out.print("Enter Name: ");
            String name = scanner.nextLine();
```

```

System.out.print("Enter Seat No: ");
String seatNo = scanner.nextLine();

System.out.print("Enter Date (DD-MM-YYYY): ");
String date = scanner.nextLine();

System.out.print("Enter Centre Number: ");
String centre = scanner.nextLine();

System.out.print("Enter marks for subject 1: ");
int marks1 = scanner.nextInt();
validateMarks(marks1);

System.out.print("Enter marks for subject 2: ");
int marks2 = scanner.nextInt();
validateMarks(marks2);

System.out.print("Enter marks for subject 3: ");
int marks3 = scanner.nextInt();
validateMarks(marks3);

System.out.println("Result:");
System.out.println("Name: " + name);
System.out.println("Seat No: " + seatNo);
System.out.println("Date: " + date);
System.out.println("Centre Number: " + centre);
System.out.println("Marks: " + marks1 + ", " + marks2 + ", " + marks3);
} catch (MarksOutOfBoundsException e) {
    System.out.println("Marks Error: " + e.getMessage());
} finally {
    System.out.println("Result calculation completed.");
    scanner.close();
}

```

```

    }

    public static void validateMarks(int marks) throws MarksOutOfBoundsException {
        if (marks < 0 || marks > 100) {
            throw new MarksOutOfBoundsException("Marks should be between 0 and 100.");
        }
    }
}

```

OUTPUT :

```

PS C:\Users\Student\Aryan_1026\Exp 7> java ResultCalculation
Enter Name: Aryan
Enter Seat No: 12334
Enter Date (DD-MM-YYYY): 13 2 2006
Enter Centre Number: 123445
Enter marks for subject 1: 20
Enter marks for subject 2: -10
Marks Error: Marks should be between 0 and 100.
Result calculation completed.
PS C:\Users\Student\Aryan_1026\Exp 7>

```

6. Post-Experiments Exercise :

2.A bank has a rule that every account must maintain a minimum balance of ₹1000.If a withdrawal makes the balance go below ₹1000, the program should throw a custom

exception MinimumBalanceException.

- The program should handle this exception and display a proper error message.
- Otherwise, it should display the new balance.

CODE :

```

import java.util.Scanner;

// Custom exception class
class MinimumBalanceException extends Exception {
    public MinimumBalanceException(String message) {
        super(message);
    }
}

```

```

}

public class BankAccount {
    private double balance;

    public BankAccount(double initialBalance) {
        this.balance = initialBalance;
    }

    // Method to withdraw amount and check minimum balance
    public void withdraw(double amount) throws MinimumBalanceException {
        if (balance - amount < 1000) { // minimum balance condition
            throw new MinimumBalanceException("Withdrawal denied! Minimum balance of ₹1000 must be maintained.");
        } else {
            balance -= amount;
            System.out.println("Withdrawal successful. New balance: ₹" + balance);
        }
    }

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        BankAccount account = new BankAccount(5000); // initial balance

        System.out.println("Bank Account Minimum Balance Check");
        System.out.println("Initial Balance: ₹5000");

        try {
            System.out.print("Enter amount to withdraw: ₹");
            double amount = scanner.nextDouble();
            account.withdraw(amount);
        } catch (MinimumBalanceException e) {
            System.out.println("Error: " + e.getMessage());
        } catch (Exception e) {
            System.out.println("Invalid input.");
        }
    }
}

```

```
} finally {  
    System.out.println("Transaction attempt finished.");  
    scanner.close();  
}  
}  
}
```

OUTPUT :

```
PS C:\Users\Student\Aryan_1026\Exp 7> javac .\BankAccount.java  
PS C:\Users\Student\Aryan_1026\Exp 7> java BankAccount  
Bank Account Minimum Balance Check  
Initial Balance: ₹5000  
Enter amount to withdraw: ₹5000  
Error: Withdrawal denied! Minimum balance of ₹1000 must be maintained.  
Transaction attempt finished.  
PS C:\Users\Student\Aryan_1026\Exp 7>
```